

**PREDICTIVE ANALYTICS IN HOSPITALITY:
A DECISION TREE APPROACH TO FORECASTING
HOTEL BOOKING CANCELLATIONS USING R**

PREDICTIVE ANALYTICS IN HOSPITALITY: A DECISION TREE APPROACH TO FORECASTING HOTEL BOOKING CANCELLATIONS USING R

Project submitted to ST. Teresa's College (Autonomous), Ernakulam,

Affiliated to Mahatma Gandhi University in partial completion of

PGDM-BUSINESS ANALYTICS

Submitted by

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Under the supervision and guidance of

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Verified
[Signature]
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CERTIFICATE

This is to certify that the project entitled **“PREDICTIVE ANALYTICS IN HOSPITALITY: A DECISION TREE APPROACH TO FORECASTING HOTEL BOOKING CANCELLATIONS USING R”** is a Bonafide record of the project work carried out by **Ms. KRISHNA RAMESH (Reg No: SM24PGDM009)** final year student of **PGDM-BUSINESS ANALYTICS** under my supervision and guidance during 2025- 2026. The project report represents the work of the candidate and is hereby approved for submission.

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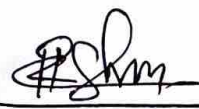
DR. SUNITHA T.R

DECLARATION

I hereby declare that the project work entitled “**PREDICTIVE ANALYTICS IN HOSPITALITY INDUSTRY: A DECISION TREE APPROACH TO FORECASTING HOTEL BOOKING CANCELLATIONS USING R**” submitted to **ST. TERESA’S COLLEGE (AUTONOMOUS), ERNAKULAM** is a record of an original work done by me under the guidance of **DR. SUNITHA T.R, Asst. Professor, ST. Teresa’s College, Ernakulam**, and this project work is submitted in the partial fulfilment of the requirements for the award of the degree of **PGDM – Business Analytics**. The results embodied in this project have not been submitted to any other University or Institute for the award of any degree in diploma.

Place : Ernakulam

Date : 31/10/25


KRISHNA RAMESH

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EXECUTIVE SUMMARY

The hospitality industry faces significant challenges due to frequent hotel booking cancellations, which directly affect revenue management, room utilization, and operational planning. The purpose of this study is to develop a Decision Tree model in R programming to predict the likelihood of hotel booking cancellations using historical reservation data. By identifying patterns and key drivers influencing cancellations, hotels can take preventive actions to minimize losses and improve customer experience.

The study utilizes decision tree modelling in R programming to analyse a dataset containing variables such as market segment type, lead time, family size, and car parking space availability. Decision trees are selected as the analytical tool due to their ability to handle both categorical and numerical data, their interpretability, and their resilience to outliers. They effectively split the data into distinct decision rules, offering a clear understanding of how various factors contribute to booking outcomes.

The study employed Classification and Regression Tree (CART) methodology through the rpart package in R, chosen for its interpretability and robustness to non-linear relationships. The analysis reveals that online bookings and longer lead times are the strongest predictors of cancellations, while larger family sizes and shorter lead times tend to reduce cancellation likelihood. These insights allow hotel managers to design targeted strategies, such as requiring deposits for high-risk bookings, offering flexible cancellation policies, and improving communication with customers.

Although the model provides reliable predictions, it does not account for external factors such as seasonal demand shifts, market trends, or customer sentiment changes, which may influence cancellation behaviour. Nonetheless, the decision tree model provides a practical and transparent approach for understanding customer booking patterns and supports better forecasting and operational decision-making in the hotel industry.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In today's competitive hospitality environment, data-driven decision-making has become essential for success. Hotels are increasingly depending on analytics to understand customer behaviour, manage operations, and make informed business decisions. One of the most pressing challenges faced by the industry is the issue of booking cancellations, which significantly impact occupancy rates, revenue management, and forecasting accuracy. When customers cancel their bookings, hotels not only lose potential revenue but also face difficulties in managing room availability and operational costs. Hence, being able to predict booking cancellations accurately can help hotels take proactive steps to minimize losses and improve efficiency.

The rise of online booking platforms and flexible cancellation policies has changed customer behaviour dramatically. Many travellers now book rooms well in advance, often with multiple options, and later cancel without financial penalties. This trend has led to increasing unpredictability for hotel managers, making it harder to plan resources such as staffing, housekeeping, and inventory management. Cancellations can also create a ripple effect, affecting marketing efforts, customer satisfaction, and brand image. As a result, hotels must seek innovative, data-backed approaches to anticipate which bookings are most likely to be cancelled and adapt their operational and marketing strategies accordingly.

The application of predictive analytics and machine learning techniques offers a promising solution to this problem. Among these, decision tree modeling stands out due to its simplicity, interpretability, and strong predictive power. Decision trees allow analysts to break down complex data into clear, rule-based structures that show how different factors contribute to a particular outcome. In the context of hotel bookings, decision trees can help identify patterns and relationships among variables such as lead time, market segment, family size, and car parking space availability, which collectively influence a customer's likelihood to cancel a booking. The visual nature of the tree also makes the insights easy for hotel managers to understand and apply in practice.

Decision trees are particularly effective because they can handle both categorical and

numerical variables without requiring complex transformations. They are also robust to outliers and missing values, making them suitable for real-world hotel datasets that often contain inconsistencies. Furthermore, unlike many statistical models, decision trees do not assume a specific data distribution, which enhances their flexibility. This feature is especially useful in the hospitality sector, where customer behaviour is often unpredictable and influenced by multiple external factors, such as seasonality, travel restrictions, or economic conditions.

This study aims to develop a decision tree model in R programming to predict whether a hotel booking will be cancelled or confirmed. The model uses real-world data from hotel bookings and incorporates variables that represent customer demographics, booking details, and stay characteristics. By analysing these features, the model seeks to identify the strongest predictors of cancellation behaviour. The outcomes will help hotel managers design preventive strategies such as offering incentives for confirmed bookings, implementing deposit-based systems, or sending timely reminders to customers with long lead times. These insights can lead to more stable occupancy rates and improved financial performance.

Beyond its practical benefits, this research also contributes to the broader understanding of customer decision-making behaviour in the hospitality industry. It highlights how predictive modeling can enhance operational planning and customer relationship management. The findings may also serve as a reference for developing similar models in related sectors, such as airlines or travel agencies, where cancellations and no-shows pose similar challenges. Overall, this project emphasizes the growing importance of integrating analytics into hotel management, demonstrating that data-driven insights are key to reducing uncertainty, improving service quality, and ensuring long-term competitiveness in the evolving global hospitality market.

1.2 STATEMENT OF PROBLEM

Despite the hotel's ongoing efforts to manage reservations efficiently through online platforms and customer engagement strategies, there remains a significant challenge in addressing the high rate of booking cancellations. The unpredictability of customer behavior particularly in online bookings with flexible cancellation policies has made it increasingly difficult for hotels to maintain stable booking levels and accurately anticipate guest arrivals. The primary

objective of this project is to develop a decision tree model that can accurately predict whether a hotel booking is likely to be canceled or not, based on variables related to customer demographics, booking details, and stay characteristics. By identifying the most influential factors that drive cancellation behavior, such as lead time, market segment, family size, and car parking space availability, the model aims to provide actionable insights into guest decision-making patterns.

1.3 SIGNIFICANCE OF THE STUDY

This research holds substantial academic and practical significance. From an academic standpoint, it illustrates how operational issues in the hospitality sector can be resolved using predictive analytics. Decision tree modeling contributes to data-driven research in hotel management by offering a visual, interpretable method of determining which factors have the greatest impact on cancellations. Moreover, it bridges the gap between theoretical data mining concepts and their practical implementation, encouraging further academic exploration of data-driven decision-making in service-oriented industries.

From a managerial standpoint, the study aids hotels understand customer behavior and putting proactive measures into place to lower cancellation rates. Managers can benefit from insights like online segment bookings indicating higher risk and long lead time bookings being more likely to be canceled. This can help them set up confirmation or reminder procedures for reservations that carry a high risk and create flexible deposit and price guidelines. Implementing such data-driven strategies not only enhances revenue stability and forecasting accuracy but also improves customer satisfaction by ensuring smoother reservation experiences.

1.4 SCOPE OF THE STUDY

The study focuses on the hotel reservations, and the factors that affect the likelihood that a reservation will be confirmed or cancelled. It is based on a real-world hotel booking dataset that includes both cancelled and non-cancelled reservations. The study uses R programming and decision tree modelling to forecast, whether a hotel reservation would be cancelled or not based on each reservation's unique attributes. The model examines a dataset

that contains variables such as market segment, lead time, number of family members, availability of parking spaces, and other booking parameters.

To provide a comprehensive knowledge of the causes that lead to cancellations, the scope of research includes various dimensions of booking data, including client type (online or offline), duration between booking and check-in, and room preferences. Decision tree analysis is selected over other statistical models such as logistic regression due to its reliability in handling non-linear relationships and outliers. Outliers might appear in real-world hotel statistics due to odd booking schedules, last-minute cancellations, or large reservations. By splitting the data into separate branches according to decision rules, decision trees can naturally handle these variances and prevent extreme values from skewing the model's overall performance.

Another benefit of using decision tree is their flexibility in processing both categorical and numerical data without the need for transformation. This makes it appropriate for the hospitality dataset, which includes a variety of variable types, including family size (discrete), lead time (numerical), and booking channels (categorical). Additionally, decision trees are simple to grasp and display, which helps decision-makers and hotel managers identify the elements that have the biggest impact on cancellations.

Although the conclusions are primarily relevant to hotel operations, they can also be applied to other hospitality-related industries that deal with internet booking trends and cancellation problems, such as resorts, homestays, and travel agencies. However, qualitative elements that can also affect cancellations, such customer satisfaction or service quality, are not included in the study. In conclusion, decision tree modeling offers a reliable and interpretable framework for analyzing complex hotel booking data.

1.5 OBJECTIVES OF THE STUDY

1. To develop a decision tree model using R that can accurately predict whether a hotel booking will be canceled or not, based on customer and booking-related attributes.
2. To analyze the influence of different variable and identify the most influential variables that contribute to booking cancellations.

3. To evaluate the model's performance and reliability using metrics such as accuracy, sensitivity, specificity, and precision derived from a confusion matrix and thereby determining its effectiveness in forecasting booking cancellations.
4. To provide practical recommendations for hotel management to minimize cancellations based on the analytical results.

1.6 METHODOLOGY

1.6.1 RESEARCH DESIGN

This research adopts a quantitative research design using secondary data and machine learning-based classification methods. A Decision Tree algorithm was chosen due to its interpretability and ability to handle both numerical and categorical variables without requiring transformation.

1.6.2 DATA SOURCE AND METHOD OF COLLECTION

The dataset was retrieved from secondary source - Kaggle website, a web-based platform, and data science community for machine learning professionals. It includes reservation details such as the number of nights, room type, lead time, and market segment along with the booking status (Canceled or Not Canceled) which makes it ideal for building a classification model.

1.6.3 DATA ATTRIBUTES

The dataset includes one dependent variable (DV) and multiple independent variables (IVs).

- **Dependent Variable (DV):**
Booking Status — a binary variable indicating whether a booking was canceled or not canceled.
- **Independent Variables (IVs):**
The independent variables include a range of booking and customer-related attributes such as

lead time (number of days between booking and arrival), market segment type (e.g., online, offline, corporate), number of adults and children, car parking space availability, room type, and special requests.

1.6.4. DATA PREPROCESSING

1. Removal of irrelevant variables (e.g., Booking ID, date of reservation).
2. Handling Missing Values

The dataset was checked using `is.na()` function (no missing values found). There were no missing values, hence additional cleaning was not required.

3. Balancing the dataset by reducing not-canceled bookings to avoid bias.

The original dataset exhibited a class imbalance, with approximately 24,000 “Not Canceled” bookings and only about 11,000 “Canceled” bookings. a data balancing technique was applied, wherein a portion of the “Not Canceled” observations was removed to avoid the risk of bias.

4. Data formatting(<chr> to <fct>)

The variables which were in ‘character’ datatype (meal type, room type, market segment type, booking status) were formatted to ‘factor’ datatype for modeling.

5. Splitting data into training (80%) and testing (20%) subsets using the `createDataPartition()` function from the `caret` package.

1.6.5. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was conducted to understand data distribution, detect outliers, and visualize variable relationships. Graphical techniques such as histograms, boxplots, and bar charts were used to explore numeric and categorical variables like lead time, room type, and market segment type. Descriptive statistics were used to summarize the key characteristics of the dataset. Measures of central tendency such as the mean, median, and mode, along with measures of dispersion were calculated for numerical variables.

1.6.6 MODEL DEVELOPMENT

A Decision Tree model was developed using the R programming language to predict the likelihood of hotel booking cancellations. The model used Booking Status as the dependent variable and all other attributes as independent variables. The resulting model was visualized using the `rpart.plot()` function for interpretation.

1.6.7 MODEL EVALUATION

The model was evaluated using a confusion matrix, which compares predicted and actual values. Metrics calculated include Accuracy, Precision, Sensitivity (Recall), Specificity, and F1 score to assess classification performance.

1.6.8 DATA ANALYSIS AND INTERPRETATION

The developed decision tree model was used to derive insights and interpret patterns in hotel booking behavior. The tree splits and confusion matrix were analyzed to derive business insights and managerial recommendations for reducing cancellations.

1.7 LIMITATIONS OF THE STUDY

- The study uses a single algorithm (Decision Tree) for prediction without comparing performance to other models such as Random Forest or Logistic Regression.
- The model was developed using a limited set of variables available in the dataset. There could be other variables that were not included but could significantly impact cancellation behavior.
- The analysis focuses mainly on quantitative aspects. Certain qualitative variables like Customer Satisfaction, reason for travel were not included which can also influence cancellations.
- Since the model was trained on historical data, its insights and predictions may not remain consistently valid in the future since the industry is dynamic and customer behaviors are constantly evolving.

CHAPTER 2

INDUSTRY

2.1 INDUSTRY PROFILE – HOSPITALITY

The Hospitality Industry is a broad sector of the service industry which primarily focus on providing welcoming services to guests, with key emphasis on customer satisfaction and creating enjoyable experience. Studies indicate that the travel and tourism industry generates \$11.1 trillion to the global GDP in 2025. The direct contribution of Indian hotel industry to the country's GDP is estimated to touch \$1 trillion by 2047 driven by a significant jump in domestic tourist visits and international arrivals. This means this industry has become the powerhouse of employment opportunities and a global contributor to the economy. The hotel industry is the core component of this sector, offering lodging and associated facilities both tourists and business travellers. In addition to providing lodging, hotels play a significant role in promoting travel by enhancing visitors' overall experiences through ease, comfort, and high-quality service. Customer contentment, excellent service, and the capacity to quickly adjust to shifting market trends are all critical to the hotel industry's success.

In recent years, the hospitality industry has undergone a significant transformation driven by digitalization and online booking systems. The way that consumers look for, evaluate, and book lodging has been completely transformed by the emergence of online travel agencies (OTAs), hotel websites, and booking platforms like Booking.com, Expedia, and Airbnb. This digital transformation has brought increased competition and transparency but has also brought challenges, such as higher rates of booking cancellations due to flexible policies and multiple booking options. Hotels must now deal with varying demand and unpredictable occupancy levels as guests may now make reservations instantaneously and cancel with no penalty. Therefore, operational effectiveness and revenue stability now depend on knowing and forecasting client booking behaviour.

Furthermore, the hospitality sector is changing quickly as a result of technological advancements and customer-focused developments. Automation, machine learning, and artificial intelligence have all been used to improve decision-making, streamline operations, and improve personalization. Hotel management can

manage overbookings, predict cancellations, and create adaptable offers that suit customer preferences with the use of tools like decision trees and predictive models. At the same time, the industry also must deal with issues including preserving service quality, controlling operating expenses, and guaranteeing long-term growth. All things considered, the hospitality sector remains a vital component of the world economy, driven by innovation, client interaction, and the never-ending quest for service excellence.

2.2 TECHNOLOGIES IN THE HOSPITALITY INDUSTRY

The past few years have seen a significant technological revolution in the hospitality sector, fuelled by rising demands for efficiency, convenience, and personalization. Nowadays, technology is crucial for hotels' operations, guest management, and delivering unforgettable experiences. The use of digital payment gateways and online booking platforms, which enable users to easily make reservations and conduct transactions from any location, has been one of the biggest developments. Hotel reservations are now quicker and more transparent because to the growth of mobile applications and online travel agencies (OTAs) like Booking.com and Expedia. These companies provide clients with instant confirmations, flexible booking options, and reviews.

To efficiently manage front-office operations, guest preferences, and communication, hotels have also embraced Property Management Systems (PMS) and Customer Relationship Management (CRM) software. Personalized recommendations, dynamic pricing, and predictive insights into consumer behavior like determining the probability of cancellations are all made possible by the growing integration of artificial intelligence (AI) and machine learning (ML) technologies. Big Data Analytics also gives hotels the ability to estimate demand and optimize resources by analyzing enormous volumes of operational and booking data.

However, the problem of privacy also arises because of increased digitization. Ensuring cybersecurity and adherence to privacy standards has grown crucial as hotels gather and preserve sensitive consumer data. Despite these obstacles, technological integration is redefining hospitality operations to become more intelligent, effective, and customer-focused.

2.3 CHALLENGES AND OPPORTUNITIES IN THE INDUSTRY

The environment in which the hotel sector operates is characterized by constant change and unpredictability. The increased number of cancellations, particularly from online platforms that provide flexible return policies, is one of the biggest problems hotels are currently facing. These cancellations cause inefficiencies in personnel, room management, and cost control by interfering with revenue forecasts and operational planning. Additionally, consumer demand and occupancy rates are greatly impacted by external factors such as seasonal variations, travel limitations, pandemics, and economic volatility. Increased reliance on digital technologies also brings with it cybersecurity threats and the requirement to safeguard client information against possible intrusions.

Despite these challenges, the sector presents countless chances for development and expansion by fusing data analytics with cutting-edge technologies. Hotels can anticipate customer demands, personalize experiences, and strengthen guest connections by utilizing AI-powered systems and predictive models. The emergence of eco-friendly practices and sustainable tourism offers hotels the chance to stand out in a market that is becoming more and more conscious. Additionally, real-time analysis of booking patterns and consumer feedback enables hotels to continuously enhance operational strategy and service quality.

Through the proper use of analytics and technology, the hospitality sector may transform obstacles into competitive advantages. Hotels can minimize uncertainty, boost productivity, and provide individualized guest experiences through data-driven decision-making, which eventually promotes long-term growth and competitiveness in a world market that is changing quickly.

2.4 APPLICATION OF ANALYTICS IN THE HOSPITALITY INDUSTRY

The hospitality sector has adopted analytics across various operational domains to enhance decision-making and competitiveness. The following are key areas where data analytics is transforming hotel management:

2.4.1 REVENUE MANAGEMENT AND FORECASTING

Revenue management involves optimizing room pricing and inventory to maximize profitability. Predictive analytics enables dynamic pricing strategies by forecasting demand, occupancy rates, and expected cancellations.

By integrating models like Decision Trees, Regression, and Time Series Forecasting, hotels can predict demand patterns, adjust room rates in real time, and minimize revenue loss caused by unexpected cancellations.

2.4.2 CUSTOMER SEGMENTATION AND BEHAVIOR ANALYSIS

Based on demographics, booking patterns, and preferences, analytics assist hotels in grouping patrons into useful segments. Personalized offers, loyalty plans, and targeted promotions are made possible by segmentation.

For instance, hotels can develop unique pricing and cancellation policies by recognizing that leisure travelers cancel more frequently than business travelers.

2.4.3 PREDICTIVE ANALYTICS FOR CANCELLATION MANAGEMENT

Booking cancellations are increasingly being predicted using predictive algorithms like Random Forests and Decision Trees. These models examine past data to determine which factors such as lead time, booking channel, and payment method have the biggest effects on cancellation decisions.

Hotels can lower operational uncertainty and increase occupancy accuracy by anticipating cancellations and putting preventive measures in place, such as overbooking regulations, deposit requirements, and automated reminder systems.

2.4.4 IMPORTANCE OF DATA-DRIVEN DECISION-MAKING IN HOSPITALITY

The way hotels organize, promote, and manage their services has been completely transformed by the incorporation of data analytics. Data-driven systems guarantee efficiency and competitiveness in everything from optimizing back-end operations to improving visitor experiences through tailored recommendations.

By identifying high-value clients, anticipating cancellations before they happen, and customizing services to optimize happiness and income, predictive modelling helps hotels make decisions that are less reactive and more proactive.

CHAPTER 3

THEORETICAL FRAMEWORK

3.1 DECISION TREE-MEANING

A decision tree is a structure that resembles a flowchart and is used to make predictions or choices. It is composed of nodes that indicate choices or attribute tests, branches that show how these choices turned out, and leaf nodes that provide the results or forecasts. Each internal node signifies a test based on an attribute, each branch indicates the result of that test, and each leaf node denotes either a class label or a continuous value.

Decision trees represent choices and their outcomes in a tree-like structure, with nodes representing events or choices and edges representing decision rules or conditions. They are frequently used to make judgments based on predictor variables in data mining and machine learning using the R programming language. Multiple algorithms are used to determine when to split a node into sub-nodes, increasing their homogeneity and purity with respect to the target variable. The decision tree chooses the split that produces the most homogenous sub-nodes by dividing nodes according to the factors that are available.

A decision tree visually represents the potential results of a sequence of interconnected decisions. It enables a person or group to compare potential actions by considering their expenses, likelihoods, and advantages. They can be utilized for either facilitating casual conversation or for outlining a mathematical algorithm that foresees the optimum selection.

It usually begins with one node that splits into various potential results. Each of these leads to further nodes, which in turn branch out into several other possible fates. As a result, a tree-like form is produced.

3.1.1 TERMINOLOGIES OF DECISION TREE

- a. Root Node: The root node of the tree symbolizes the whole population or sample and can be divided into two or more similar sets.
- b. Splitting: The action of dividing a node into two or more sub-nodes is referred to as splitting.
- c. Branch: A branch is a segment of the complete tree structure.
- d. Decision Node: A decision node is formed when a sub-node further divides into additional sub-nodes.
- e. Parent Node: The parent node serves as the origin for sub-nodes that branch out from it, making them the children of the parent node.
- f. Leaf Node: A node that does not undergo further division is referred to as a leaf or terminal node.

- g. Pruning: Pruning is the process of eliminating sub-nodes from a decision node, acting as the reverse of splitting.

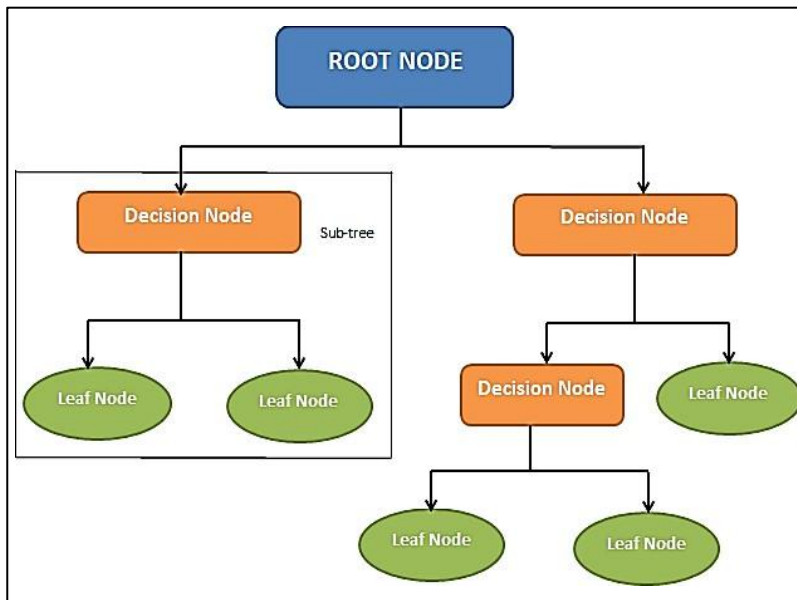


Figure 1

Decision tree algorithm

3.2 CART ALGORITHM

The CART Algorithm is a decision tree technique applicable to both classification and regression problems. It is a supervised learning algorithm that leverages labeled data to make predictions on unseen data.

CART is a general term that encompasses the following types of decision trees:

- **CLASSIFICATION TREES:** These trees are utilized to identify the most probable "class" into which the target variable falls when it is continuous.
- **REGRESSION TREES:** These trees are used to predict the value of a continuous variable.

CART creates a hierarchical structure that consists of nodes and branches. The nodes signify points of decision, while the branches illustrate the possible results of those choices. The terminal nodes of the tree contain a predicted class

label or value for the dependent variable.

For data splitting, CART adopts a greedy approach where it assesses all feasible splits and chooses the one that best diminishes the impurity of the resulting subsets. In classification tasks,

CART utilizes Gini impurity as its criterion for splitting, seeking to minimize Gini impurity to create more homogenous subsets. In regression tasks, CART applies residual reduction as the criterion for splitting, focusing on reducing residuals to enhance the model's accuracy in representing the data.

To prevent overfitting, pruning is employed to eliminate nodes that contribute minimally to the model's accuracy. Cost complexity pruning and information gain pruning are two well-known pruning techniques. Cost complexity pruning calculates the cost of each node and removes nodes with negative cost. Information gain pruning calculates the information gain of each node and removes nodes with low information gain.

3.2.1 CART FOR CLASSIFICATION

A classification tree is a technique used to determine the most probable "Class" of the categorical target variable. It is applied when there is a need to categorize the dataset into classes linked to the response variable, such as "yes" or "no."

In the context of decision tree learning for classification, a specific algorithm is used to build a tree structure aimed at forecasting class labels. This tree includes nodes that represent various decision points and branches that show the possible outcomes of those decisions. The class labels that are predicted are located at each leaf node of the tree. In this study, CART for Classification is employed for the analysis because the response variable is categorical (Canceled or Not Canceled).

3.3 VARIABLE IMPORTANCE AND FEATURE SELECTION

Feature selection is a crucial step in predictive analytics, as irrelevant or redundant variables can reduce model accuracy. In this project, feature selection was carried out using both domain

knowledge and statistical reasoning.

The initial dataset contained 17 attributes, but after analysis, only 10 relevant variables were retained. These were chosen based on their contribution to predicting booking cancellations and removing noise from the dataset.

The key predictor variables identified by the Decision Tree were:

- Market Segment Type (Online vs. Offline)
- Lead Time
- Family Size
- Car Parking Space
- Room Type

Each of these variables provides behavioural insights:

- Online bookings indicate a higher cancellation rate due to ease of modification.
- Longer lead times show less commitment and higher uncertainty.
- Family bookings are less likely to cancel as travel plans are usually fixed.
- Parking space requests often imply local guests with higher stay likelihood.

This variable importance analysis aids in developing targeted hotel policies and improving predictive performance.

3.4 CUSTOMER SEGMENTATION AND BEHAVIORAL INSIGHTS

Customer segmentation plays a critical role in applying predictive models to business strategy. In the hospitality sector, segmentation helps classify guests based on booking behavior, preferences, and cancellation patterns.

Using the Decision Tree model's splits, distinct customer segments can be identified:

Segment 1: Online, Long Lead-Time Guests — Highest cancellation probability.

Segment 2: Offline, Short Lead-Time, Family Guests — Least likely to cancel.

Segment 3: Short Lead-Time, Premium Room Type — Moderate cancellation likelihood but high value.

Such segmentation enables hotels to adopt differentiated strategies for example, requiring partial payment for high-risk segments, or offering loyalty benefits for low-risk guests.

Predictive segmentation not only minimizes cancellation losses but also strengthens customer retention and long-term profitability. By combining machine learning with domain

understanding, hotels can shift from reactive cancellation management to proactive planning.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

1. DATA ANALYSIS

4.1.1 DATASET

- Dataset name: Booking Dataset
- Description: The dataset used in this study is available in .csv format and contains information related to hotel bookings. The data includes details about customer reservations, stay duration, booking channels, and other related features. Each observation in the dataset represents a single booking record with variables describing both customer characteristics and booking-related attributes. The objective of the dataset is to analyse and predict whether a booking will be cancelled or not cancelled based on these factors.
- Attributes: The raw dataset includes several variables about each booking. These include customer-related attributes as well as booking characteristics. The attributes are shown below in the following table.

Table 1

Attributes of the dataset

File format:	CSV format
Data format:	Structured
Number of columns:	17
Number of rows:	36285
N/A values:	Nil

- Source: The dataset was retrieved from Kaggle, a web-based platform and data science community for machine learning professionals.
- Overview of the dataset

	Booking_ID	number of adults	number of children	number of weekend nights	number of week nights	type of meal	car parking space	room type	lead time	market segment type
1	INN00001	1	1	2	5	Meal Plan 1	0	Room_Type 1	224	Offline
2	INN00002	1	0	1	3	Not Selected	0	Room_Type 1	5	Online
3	INN00003	2	1	1	3	Meal Plan 1	0	Room_Type 1	1	Online
4	INN00004	1	0	0	2	Meal Plan 1	0	Room_Type 1	211	Online
5	INN00005	1	0	1	2	Not Selected	0	Room_Type 1	48	Online
6	INN00006	1	0	0	2	Meal Plan 2	0	Room_Type 1	346	Offline
7	INN00007	1	1	1	4	Meal Plan 1	0	Room_Type 1	34	Online
8	INN00008	3	0	1	3	Meal Plan 1	0	Room_Type 4	83	Online
9	INN00009	1	1	0	4	Meal Plan 1	0	Room_Type 1	121	Offline
10	INN00010	2	0	0	5	Meal Plan 1	0	Room_Type 4	44	Online
11	INN00011	1	0	1	0	Not Selected	0	Room_Type 1	0	Online
12	INN00012	1	0	2	1	Meal Plan 1	0	Room_Type 4	35	Online
13	INN00013	2	0	2	1	Not Selected	0	Room_Type 1	30	Online
14	INN00014	1	0	2	0	Meal Plan 1	0	Room_Type 1	95	Online
15	INN00015	2	0	0	2	Meal Plan 1	0	Room_Type 1	47	Online
16	INN00016	2	0	0	2	Meal Plan 2	0	Room_Type 1	256	Online
17	INN00017	1	0	1	0	Meal Plan 1	0	Room_Type 1	0	Offline
18	INN00018	2	0	1	3	Not Selected	0	Room_Type 1	1	Online
19	INN00019	2	0	2	2	Meal Plan 1	0	Room_Type 1	99	Online

Showing 1 to 19 of 36,285 entries, 17 total columns

Figure 2
Overview of the dataset

4.1.2 INSTALLATION OF PACKAGES

In R, there are multiple packages available for performing decision tree analysis. These packages offer features for creating, displaying, and assessing decision trees. The packages which are needed to be installed for the analysis in R are:

1. tidyverse

The 'tidyverse' is a set of packages that work in harmony because they share common data representations and 'API' design. This package is designed to make it easy to install and load multiple 'tidyverse' packages in a single step.

2. dplyr

A powerful tool for data transformation and manipulation in R is the dplyr package. It offers a number of features that make data manipulation jobs easier to complete quickly and intuitively.

3. rpart

This is one of the most popular packages used for creating decision trees.

4. **rpart.plot**

Plot an rpart model, automatically tailoring the plot for the model's response type. It is an extension of the rpart package.

5. **caret**

The caret package offers a consistent way to train different models, such as decision trees, with support for model training, tuning, and evaluation.

6. **caTools**

The R package caTools is specifically created for the analysis of categorical data, with a focus on managing categorical variables and simplifying the process of generating decision trees.

R CODE

```
install.packages("tidyverse")
install.packages("dplyr")
library(tidyverse)
library(dplyr)

install.packages("rpart")
library(rpart)
install.packages("rpart.plot")
library(rpart.plot)
install.packages("caret")
library(caret)
install.packages("caTools")
library(caTools)
```

Figure 3

R code of installing packages

CODE EXPLANATION

- `install.packages("tidyverse")`
 - installing tidyverse package in R

- `library(tidyverse)`
 - loading the package in R
- `install.packages("dplyr")`
 - installing dplyr package in R
- `library(dplyr)`
 - loading the package in R
- `install.packages("rpart")`
 - install rpart R package
- `library(rpart)`
 - loading rpart libraries
- `install.packages("rpart.plot")`
 - install rpart.plot package in R
- `library(rpart.plot)`
 - loading rpart.plot libraries
- `install.packages("caret")`
 - install caret package in R
- `library(caret)`
 - loading caret libraries
- `install.packages("caTools")`
 - install caTools package in R
- `library(caTools)`
 - loading caTools libraries

4.1.3 DATA IMPORTING

The first step in the analysis is importing the dataset in RStudio. The

following R code helps you to import the dataset 'Booking.csv'

R CODE

```
2 library(readr)
3 hotelbooking <- read_csv("C:/Users/krish/OneDrive/Desktop/R Project/booking.csv")
```

Figure 4

R code of importing the dataset

CODE EXPLANATION

- `hotelbooking<- read_csv("booking.csv")`
 - assign the CSV file to an object called 'hotelbooking'

4.1.4 DATA PREPROCESSING

The crucial part of the data analysis process is data preprocessing. The process of converting unprocessed data into a format better suited for analysis and modeling is known as data preparation. To get rid of mistakes, missing numbers, and other irregularities, data must be cleaned and arranged. The dataset I received was clean with no N/A values but minimal cleaning was required. The dataset was manually balanced (in Excel) to ensure an equal distribution of canceled and not-canceled observations, to avoid model bias and improve classification accuracy. The data was then re-structured by removing irrelevant records based on specific criteria (explained in Data Cleaning step below) to make it fit for the analysis using Decision tree in R programming language.

4.1.4.1 DATA EXAMINATION

The function 'summary' is commonly used to generate result summaries for the output of different model fitting functions in the R programming language. The summary function provides the minimum, maximum, mean, median, and 1st and 3rd quartiles for a numerical vector.

R CODE

```
16 summary(hotelbooking)
```

Figure 5

R code of data examination using summary function

CODE EXPLANATION

- `summary(hotelbooking)`
 - To summarise the dataset 'hotelbooking'.

OUTPUT

Below is the summary of the dataset:

```
> summary(hotelbooking)
Booking_ID      number of adults number of children number of weekend nights
Length:36285    Min.   :0.000      Min.   : 0.0000      Min.   :0.0000
Class :character 1st Qu.:2.000      1st Qu.: 0.0000      1st Qu.:0.0000
Mode  :character Median :2.000      Median : 0.0000      Median :1.0000
                Mean   :1.845      Mean   : 0.1054      Mean   :0.8107
                3rd Qu.:2.000      3rd Qu.: 0.0000      3rd Qu.:2.0000
                Max.   :4.000      Max.   :10.0000      Max.   :7.0000

number of week nights type of meal      car parking space room type
Min.   : 0.000      Length:36285      Min.   :0.00000      Length:36285
1st Qu.: 1.000      Class :character  1st Qu.:0.00000      Class :character
Median : 2.000      Mode  :character  Median :0.00000      Mode  :character
Mean   : 2.205                        Mean   :0.03098
3rd Qu.: 3.000                        3rd Qu.:0.00000
Max.   :17.000                        Max.   :1.00000

lead time      market segment type      repeated      P-C      P-not-C
Min.   : 0.00      Length:36285      Min.   :0.00000      Min.   : 0.00000      Min.   : 0.0000
1st Qu.: 17.00      Class :character  1st Qu.:0.00000      1st Qu.: 0.00000      1st Qu.: 0.0000
Median : 57.00      Mode  :character  Median :0.00000      Median : 0.00000      Median : 0.0000
Mean   : 85.24                        Mean   :0.02563      Mean   : 0.02334      Mean   : 0.1534
3rd Qu.:126.00                        3rd Qu.:0.00000      3rd Qu.: 0.00000      3rd Qu.: 0.0000
Max.   :443.00                        Max.   :1.00000      Max.   :13.00000      Max.   :58.0000

average price      special requests date of reservation booking status
Min.   : 0.00      Min.   :0.0000      Length:36285      Length:36285
1st Qu.: 80.30      1st Qu.:0.0000      Class :character  Class :character
Median : 99.45      Median :0.0000      Mode  :character  Mode  :character
Mean   :103.42      Mean   :0.6197
3rd Qu.:120.00      3rd Qu.:1.0000
Max.   :540.00      Max.   :5.0000

> |
```

Figure 6

Screenshot of summary of raw dataset in R

4.1.4.2 DATA CLEANING

Data cleaning is a crucial step in preparing the dataset for analysis. It involves identifying and correcting inaccurate, incomplete, or irrelevant data to ensure the reliability of the model. In this study, the cleaning process was carried out after importing the manually balanced dataset into RStudio.

The dataset initially contained several variables that were not relevant to predicting booking cancellations. These variables were removed to retain only meaningful attributes. This process ensured that unnecessary columns, such as identifiers and non-predictive fields, were excluded, keeping the dataset concise and focused for modelling purposes.

Additionally, two existing columns “Number of Adults” and “Number of Children” were combined to create a new derived variable, Family Size, representing the total number of guests in each booking. This transformation provided a more meaningful representation of guest composition and was expected to influence cancellation behaviour. After creating the new column, the dataset was reviewed to ensure there were no missing or duplicate values. The final cleaned dataset was then used for factor conversion and further analysis.

R CODE

```
9 booking1 <- hotelbooking %>%
10   mutate(family_size = `number of adults` + `number of children`)
11 view(booking1)
12
13 bookingdata1<-booking1 %>%
14   select(-Booking_ID,-repeated,-`number of adults`, -`number of children`,
15         -repeated,-`P-C`, -`P-not-C`, -`average price`, -`special requests`,
16         -`date of reservation`)
17 view(bookingdata1)
18 glimpse(bookingdata1)
```

Figure 7

R code of data cleaning

CODE EXPLANATION

- `booking1 <- hotelbooking %>%`
 - Assigns the dataset `hotelbooking` to a new object `booking1` and starts a pipeline using `%>%`.
- `mutate(family_size = number of adults + number of children)`
 - Creates a new column `family_size` by adding number of adults and number of children.
- `view(booking1)`
 - Opens the `booking1` dataset in a spreadsheet-like viewer.
- `bookingdata1 <- booking1 %>%`
 - Assigns `booking1` to a new dataset `bookingdata1` and starts another pipeline.
- `select(-Booking_ID, -repeated, -number of adults, -number of children, -P-C, -P-not-C, -average price, -special requests, -date of reservation)`
 - Removes (drops) the listed columns from the dataset.
- `view(bookingdata1)`
 - Opens the `bookingdata1` dataset in a viewer.
- `glimpse(bookingdata1)`
 - Displays a compact summary of `bookingdata1`, showing variable names, types, and sample values.

RESULT

```
> view(bookingdata1)
> glimpse(bookingdata1)
Rows: 10,688
Columns: 9
$ family_size           <dbl> 2, 2, 2, 2, 3, 2, 2, 2, 1, 2, 2, 3, 2, 3, 2, 1, 1, 2,...
$ `number of weekend nights` <dbl> 1, 0, 0, 1, 2, 0, 1, 0, 0, 0, 0, 0, 2, 0, 1, 1, 0, 2,...
$ `number of week nights` <dbl> 3, 3, 1, 2, 0, 3, 2, 1, 3, 4, 2, 3, 5, 1, 2, 1, 1, 3,...
$ `type of meal`         <chr> "Not Selected", "Meal Plan 1", "Not Selected", "Not S...
$ `car parking space`    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ `room type`            <chr> "Room_Type 1", "Room_Type 4", "Room_Type 1", "Room_Ty...
$ `lead time`            <dbl> 7, 97, 2, 77, 77, 64, 72, 12, 50, 25, 32, 104, 39, 0,...
$ `market segment type`  <chr> "Online", "Online", "Online", "Online", "Online", "On...
$ `booking status`       <chr> "Not_Canceled", "Not_Canceled", "Not_Canceled", "Not_...
```


Figure 8

Screenshot of result of glimpse in R

- The number of columns reduced from 17 to 9
- The number of rows reduced from 36285 to 10688.

4.1.4.3 DATA FORMATTING

In R, the decision tree analysis requires to convert specific columns of a data frame (bookingdata1) into factors. Factors are used in R to represent categorical data, which are particularly useful for statistical modelling and analysis. All the categorical variables are converted from character datatype to factor data type.

The categorical variables in bookingdata1 data frame are: type of meal, room type, market segment type and booking status.

R CODE

```
#categorical to factors
bookingdata1$type of meal<-as.factor(bookingdata1$type of meal)
bookingdata1$room type<-as.factor(bookingdata1$room type)
bookingdata1$market segment type<-as.factor(bookingdata1$market segment type)
bookingdata1$booking status<-as.factor(bookingdata1$booking status)
glimpse(bookingdata1)
```

Figure 9

R code of data formatting

CODE EXPLANATION

- bookingdata1\$type of meal<-as.factor(bookingdata1\$type of meal)
-Convert variable type of meal datatype from <chr> to <fct>
- bookingdata1\$room type<-as.factor(bookingdata1\$room type)
-Convert variable room type datatype from <chr> to <fct>
- bookingdata1\$market segment type<-as.factor(bookingdata1\$market segment type)

-Convert variable market segment type datatype from <chr> to <fct>

- `bookingdata1$`booking status`<-as.factor(bookingdata1$`booking status`)`

-Convert variable booking status datatype from <chr> to <fct>

OUTPUT

```
> glimpse(bookingdata1)
Rows: 10,688
Columns: 9
$ family_size      <dbl> 2, 2, 2, 2, 3, 2, 2, 2, 1, 2, 2, 3, 2, 3, 2, 1, 1, 2,...
$ `number of weekend nights` <dbl> 1, 0, 0, 1, 2, 0, 1, 0, 0, 0, 0, 0, 2, 0, 1, 1, 0, 2,...
$ `number of week nights` <dbl> 3, 3, 1, 2, 0, 3, 2, 1, 3, 4, 2, 3, 5, 1, 2, 1, 1, 3,...
$ `type of meal`    <fct> Not Selected, Meal Plan 1, Not Selected, Not Selected...
$ `car parking space` <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ `room type`      <fct> Room_Type 1, Room_Type 4, Room_Type 1, Room_Type 1, R...
$ `lead time`      <dbl> 7, 97, 2, 77, 77, 64, 72, 12, 50, 25, 32, 104, 39, 0,...
$ `market segment type` <fct> Online, Online, Online, Online, Online, Online, Onlin...
$ `booking status` <fct> Not_Canceled, Not_Canceled, Not_Canceled, Not_Cancele...
> |
```

Figure 10

Screenshot of the output of glimpse after data formatting

Now, all the variables in the dataframe 'bookingdata1' contains variables of factor and double data types fulfilling the requirement of the data analysis using decision tree.

4.1.5 SELECTION OF VARIABLES

The table below shows the list of variables, all independent variables and one dependent variable, whether it is included or excluded for the analysis and the reason behind it.

Table 2

Variable Name	Description	Status	Reason for Inclusion / Exclusion
Booking_ID	Unique identification number for each booking	Excluded	Does not influence cancellation outcome

Date of Reservation	Date on which the booking was made	Excluded	Irrelevant for prediction; removed for simplification
Number of Adults	Number of adults included in the booking	Transformed	Combined with children to form <i>Family Size</i> variable
Number of Children	Number of children included in the booking	Transformed	Combined with adults to form <i>Family Size</i> variable
Number of Weekend Nights	Number of weekend nights in the stay	Retained	Indicates booking duration and travel purpose
Number of Week Nights	Number of weekday nights in the stay	Retained	Reflects booking duration and trip pattern
Type of Meal	Type of meal plan selected by the guest	Retained	Represents customer preference and spending pattern
Car Parking Space	Indicates whether a parking space was requested	Retained	Can signal local guests and booking stability
Room Type	Category or class of room booked	Retained	Influences cancellation tendency and price sensitivity
Lead Time	Number of days between booking and arrival	Retained	Major predictor of cancellation likelihood
Market Segment Type	Source/channel through which the booking was made	Retained	Important factor (online vs. offline) for cancellations
Average Price	Average price per night	Excluded	Contained zero values

Special Requests	Number of special requests made by the guest	Excluded	Sparse data with minimal predictive relevance
Arrival Month / Year	Period of booking	Excluded	Temporal variable; not required for this model
Booking Status (DV)	Final status of the booking (Cancelled / Not_Cancelled)	Retained	Target variable for prediction

List of variables, IV/DV, included/not and the reason.

4.1.6 EXPLORATORY DATA ANALYSIS (EDA)

The next step in the data pre-processing is EDA. Exploratory Data Analysis (EDA) is an essential phase in data analytics discovers general trends in the data. It understands the dataset's structure, detect patterns, and identify any irregularities before model development. Outliers and potentially unexpected aspects of the data are captured in these patterns. It serves as the foundation for comprehending and analyzing complex data sets. It involves the use of statistical summaries and visualization techniques to gain meaningful insights into the variables and their relationships with the target variable.

In this study, EDA was carried out on the balanced hotel booking dataset consisting of 10,688 observations and 9 variables. The objective was to analyze booking behaviours and identify the factors that influence whether a booking is likely to be cancelled or not cancelled.

In R Programming Language, EDA will involve two main categories:

- Graphical Methods- encompassing histogram, density estimation, box plots, and other visual representations.
- Descriptive Statistics- encompassing mean, median, mode, inter-quartile range, and other measures.

4.1.6.1 GRAPHICAL METHOD

EDA plots created in R for data visualization and analysis under graphical method are shown below:

Table 3

Type of EDA plots used for each variable type

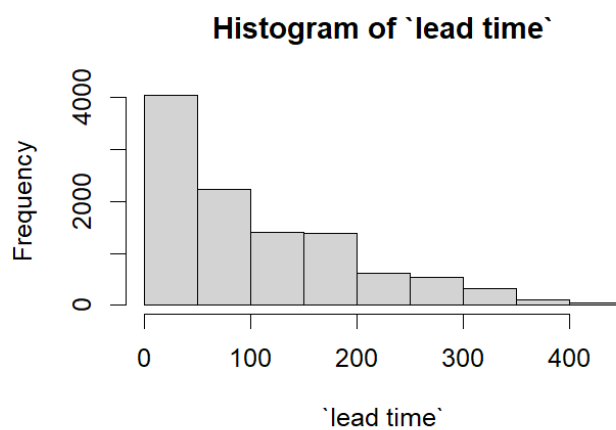
TYPE OF EDA PLOT	VARIABLE TYPE
Histogram	Numeric
Density plot	Numeric
Box plot	Numeric
Bar plot	Categorical

The following contains input variables and output variable along with their description, variable type, data type, EDA plot used, R code and the plots visualized with interpretations:

➤ **INPUT VARIABLE 1: LEAD TIME**

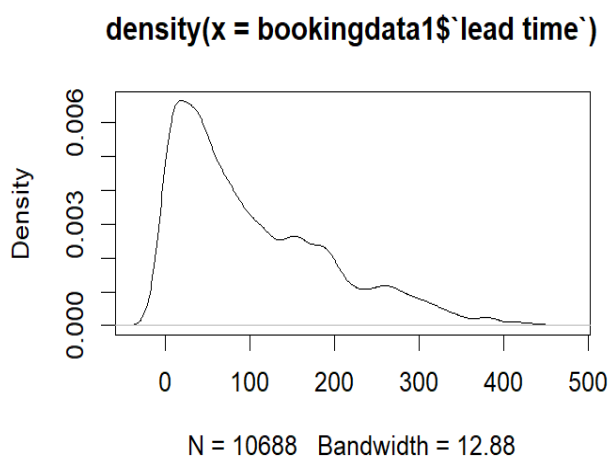
- Description: Number of days between the date of booking and the date of arrival.
- Variable type: Numeric variable
- Data type: <dbl>
- EDA plot used: Histogram, Density plot, and Box plot
- Code:
 - hist(bookingdata1\$`lead time`,
xlab = "`lead time`",main = "Histogram of `lead time`")
 - plot(density(bookingdata1\$`lead time`))
 - ggplot(data = bookingdata1, aes(x = `booking status`, y = `lead time`)) +
geom_boxplot() + labs(title = "Lead Time by Booking Status") + theme_classic()
- Plots:

Figure 11



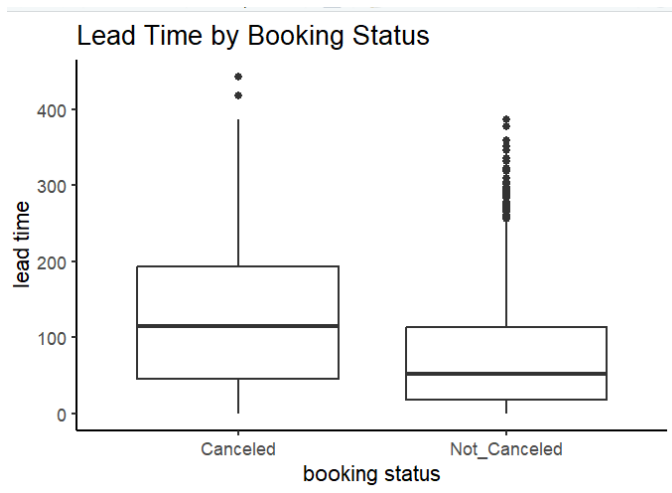
Histogram of Lead Time

Figure 12



Density plot of Lead Time

Figure 13



Boxplot of Lead Time

g. Interpretations:

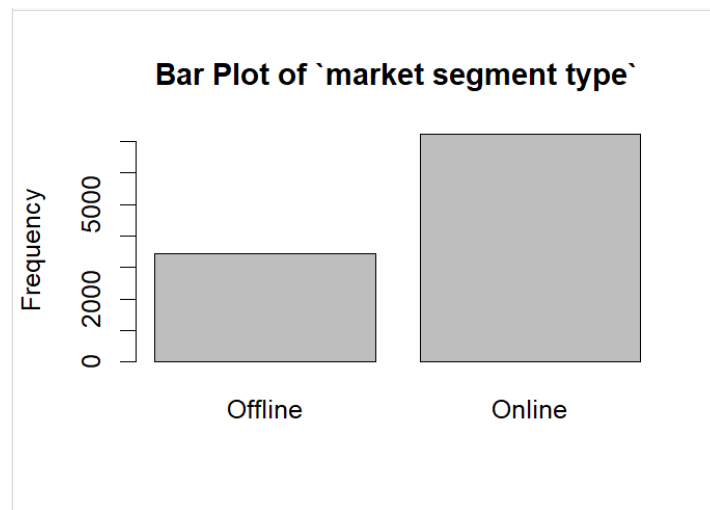
- The histogram shows that most bookings have a lead time of less than 100 days, while a few extend beyond 300 days.
- The density plot indicates a right-skewed distribution, meaning long lead-time bookings are less common but significant.
- The boxplot reveals that longer lead times are often associated with cancelled bookings, suggesting that customers booking far in advance are more likely to cancel.

➤ **INPUT VARIABLE 2: MARKET SEGMENT TYPE**

- a. Description: Booking channel used by the customer (Online or Offline)
- b. Variable type: Categorical variable
- c. Data type: <fct>
- d. EDA plot used: Bar plot
- e. R code:
 - `barplot(table(bookingdata1$`market segment type`),
main = "Bar Plot of `market segment type`",ylab = "Frequency")`

f. Plot:

Figure 14



Bar plot of Market Segment

g. Interpretation:

- The majority of bookings were made through online channels, followed by offline.
- Online bookings recorded a higher number of cancellations, likely due to flexible refund.

➤ INPUT VARIABLE 3: FAMILY SIZE

- Description: Total number of people in a booking, calculated by adding adults and children.
- Variable type: Numerical variable
- Data type: <dbl>
- EDA plot used: Histogram, Density Plot, and Box Plot
- R code:

- `hist(bookingdata1$family_Size, xlab = "family_Size", main = "Histogram of family_Size")`
- `plot(density(bookingdata1$family_Size))`
- `ggplot(data = bookingdata1, aes(x = `booking status`, y = family_Size)) + geom_boxplot() + labs(title = "Family Size by Booking Status") + theme_classic()`

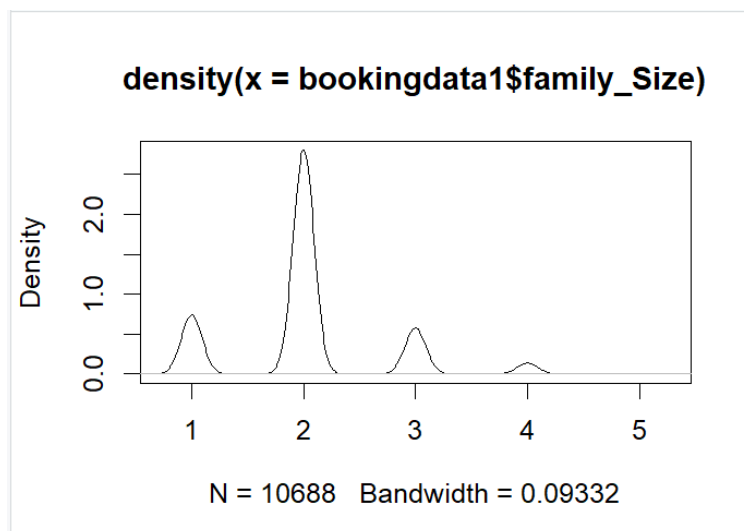
f. Plot:

Figure 15



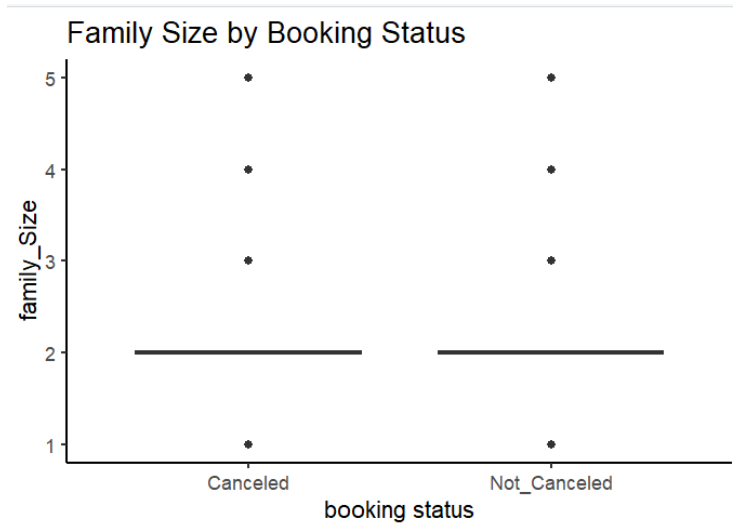
Histogram of Family Size

Figure 16



Density Plot of Family Size

Figure 17



Box Plot of Family Size

g. Interpretation:

- The histogram indicates that most bookings are made by small families (2 to 3 members).
- The density plot shows a normal distribution centered around family size = 2.
- The boxplot reveals that larger families are less likely to cancel compared to single travelers or couples, possibly due to stronger commitment to travel plans.

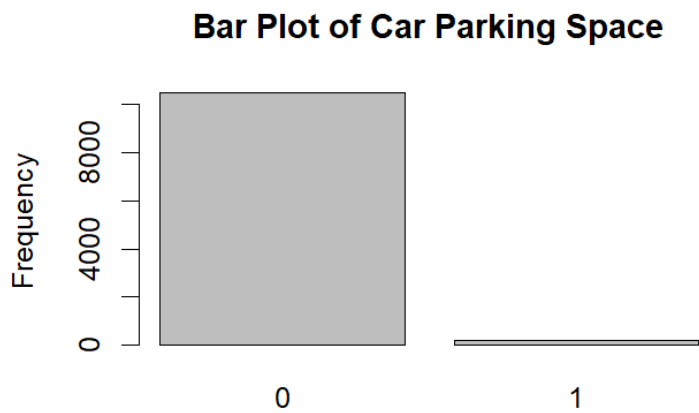
➤ INPUT VARIABLE 4: CAR PARKING SPACE

- Description: Car parking spaces requested by guests.
- Variable type: Numeric (Binary)
- Data type: <dbl>
- EDA plot used: Bar plot
- R code:

- ```
barplot(table(bookingdata1$`car parking space`),
main = "Bar Plot of Car Parking Space", ylab = "Frequency")
```

f. Plot:

**Figure 18**



*Bar plot of Car Parking Space*

g. Interpretation:

- Most bookings did not include a car parking request.
- Guests who requested parking were more likely to retain their booking, indicating that customers with planned road travel tend to cancel less.

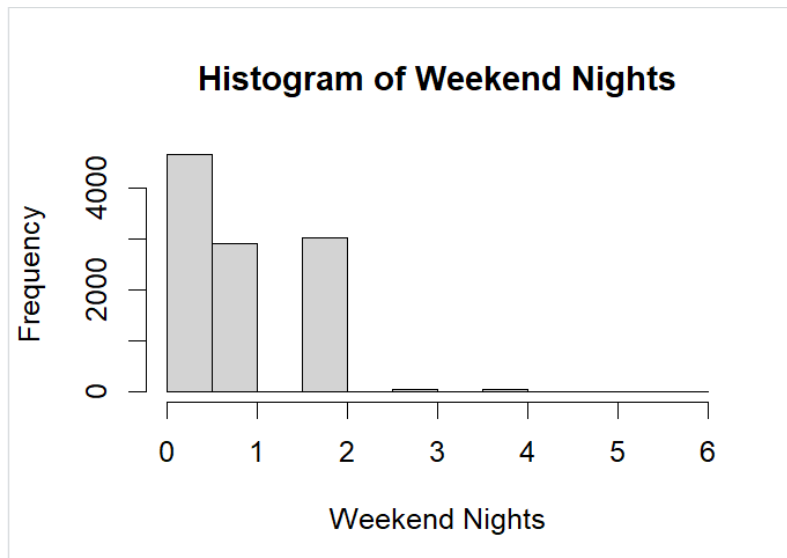
#### ➤ INPUT VARIABLE 5: NUMBER OF WEEKEND NIGHTS

- a. Description: Number of Nights stayed during weekends.
- b. Variable type: Numeric variable
- c. Data type: <dbl>
- d. EDA plot used: Histogram and Box plot
- e. R code:
  - `hist(bookingdata1$`number of weekend nights`,  
xlab = "Weekend Nights", main = "Histogram of Weekend Nights")`
  - `ggplot(data = bookingdata1, aes(x = `booking status`,  
y = `number of weekend nights`)) + geom_boxplot()`

```
+ labs(title = "Weekend Nights by Booking Status") +theme_classic()
```

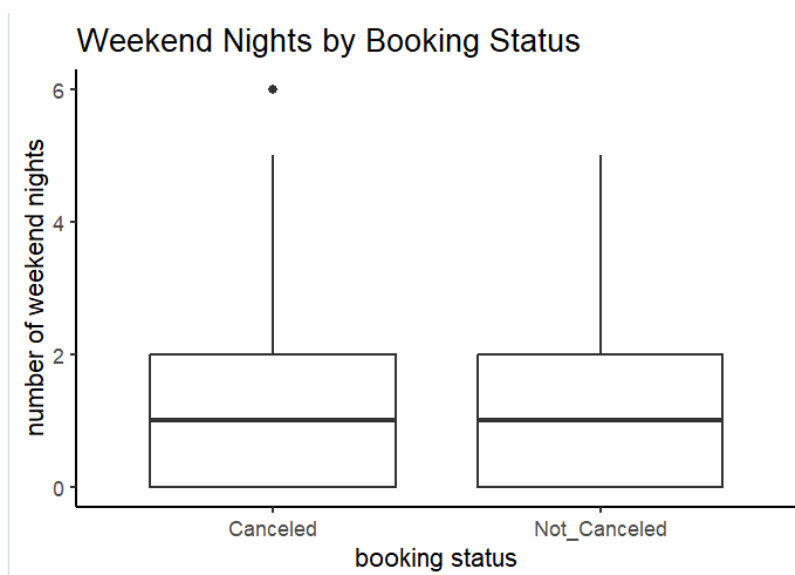
f. Plots:

**Figure 19**



*Histogram of Weekend Night*

**Figure 20**



*Box plot of Weekend Night*

g. Interpretations:

- The histogram shows that most guests stayed for one or two weekend nights.
- Boxplot analysis indicates that cancellations are higher for shorter weekend stays, suggesting that short leisure trips are more flexible and prone to changes.

➤ **INPUT VARIABLE 6: NUMBER OF WEEK NIGHTS**

a. Description: Number of nights stayed during weekdays.

b. Variable type: Numerical variable

c. Data type: <dbl>

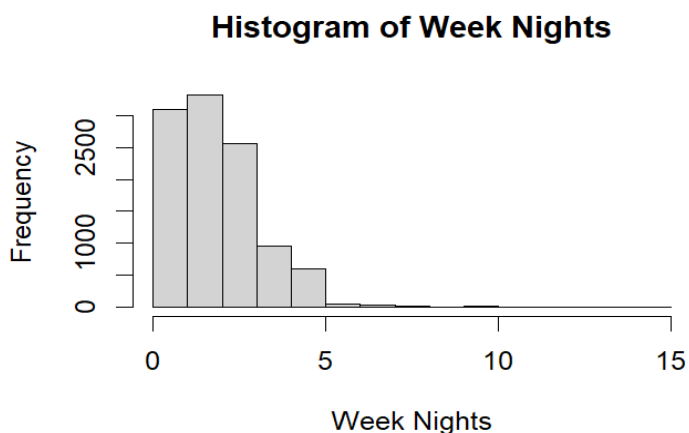
d. EDA plot used: Histogram and Box plot

e. R code:

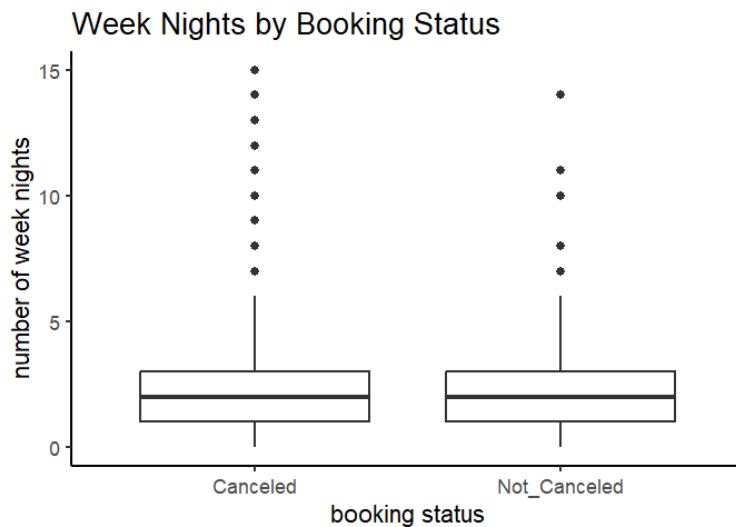
- `hist(bookingdata1$`number of week nights`,  
xlab = "Week Nights", main = "Histogram of Week Nights")`
- `ggplot(data = bookingdata1, aes(x = `booking status`, y = `number of week nights`)) +  
geom_boxplot() + labs(title = "Week Nights by Booking Status") + theme_classic()`

f. Plot:

**Figure 21 :**



**Figure 22:**



*Box Plot of Week Nights*

g. Interpretations:

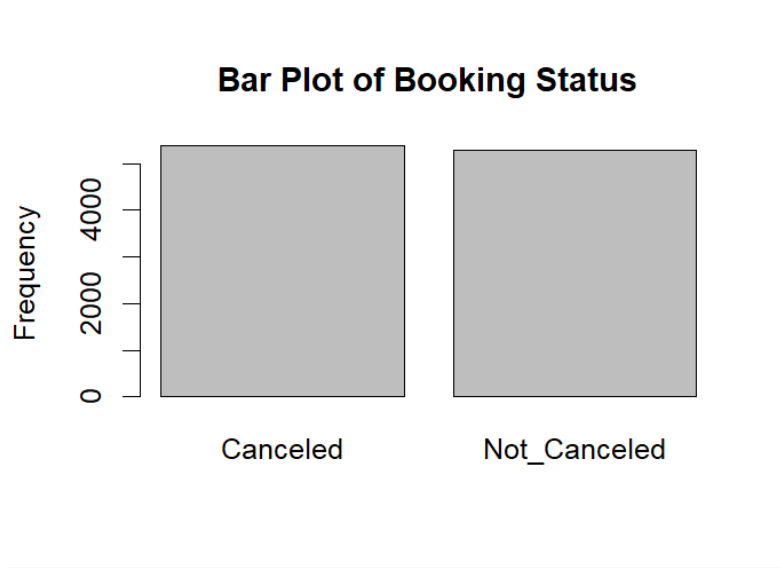
- Most guests stayed for 2–3 weekday nights.
- Bookings with longer stays showed lower cancellation rates, indicating that extended bookings tend to be more stable.

➤ **OUTPUT VARIABLE : BOOKING STATUS**

- Description: Indicates whether the booking was canceled or not.
- Variable type: Categorical variable - (binary variable- Canceled, Not Canceled)
- Data type: <chr>
- EDA plot used: Bar plot
- R code:
  - `barplot(table(bookingdata1$`booking status`),`  
`main = "Bar Plot of Booking Status",`  
`ylab = "Frequency")`

f. Plot:

**Figure 23**



*Bar plot showing Booking Cancellation*

g. Interpretations:

Initially the bar plot showed that the dataset initially contained more “Not Canceled” bookings compared to “Canceled” ones. The data was later balanced by removing some “Not Canceled” records to ensure that the model learned both classes equally. This balance improved model fairness and reduced bias during classification.

#### **4.1.6.2 DESCRIPTIVE STATISTICS**

Descriptive statistics is the second classification of performing EDA in R Programming Language. This stage provides numerical summaries of variables to describe their main characteristics in a meaningful and interpretable manner. Measures of central tendency (mean, median, mode) and dispersion (minimum, maximum, standard deviation, and interquartile range) were used to understand how data values are distributed and to detect any anomalies or outliers. Through descriptive analysis, we present our data in a meaningful manner to facilitate easy

understanding. Measures such as central tendency and variability or dispersion are employed to describe a data set.

## R CODE

```
6 summary(bookingdata1)
```

**Figure 24**

*R code of summary of the dataset*

## CODE EXPLANATION

- `summary(bookingdata1)`
- Provides descriptive statistics such as minimum, maximum, mean, and median for numerical variables, and frequency counts for categorical variables in the dataset.

## RESULT

```
> view(bookingdata1)
> summary(bookingdata1)
 family_Size number of weekend nights number of week nights type of meal
Min. :1.000 Min. :0.0000 Min. : 0.000 Meal Plan 1 :8186
1st Qu.:2.000 1st Qu.:0.0000 1st Qu.: 1.000 Meal Plan 2 : 908
Median :2.000 Median :1.0000 Median : 2.000 Not Selected:1594
Mean :2.027 Mean :0.8778 Mean : 2.332
3rd Qu.:2.000 3rd Qu.:2.0000 3rd Qu.: 3.000
Max. :5.000 Max. :6.0000 Max. :15.000

 car parking space room type lead time market segment type
Min. :0.00000 Room_Type 1:7976 Min. : 0.0 Offline:3444
1st Qu.:0.00000 Room_Type 2: 205 1st Qu.: 32.0 Online :7244
Median :0.00000 Room_Type 3: 1 Median : 76.0
Mean :0.01918 Room_Type 4:2040 Mean :103.9
3rd Qu.:0.00000 Room_Type 5: 66 3rd Qu.:160.0
Max. :1.00000 Room_Type 6: 362 Max. :443.0
 Room_Type 7: 38

 booking status
Canceled :5393
Not_Canceled:5295
```

**Figure 25**

*Screenshot of the result (summary) of the dataset*



## INTERPRETATIONS

### i. LEAD TIME

- The minimum lead time is 0 days, and the maximum is around 480 days, with a median of approximately 90 days.
- This shows that while most bookings are made within a few weeks, some guests plan several months in advance. Longer lead times tend to correlate with a higher likelihood of cancellation.

### ii. MARKET SEGMENT TYPE

- Most bookings were made through online channels, followed by offline.
- Online bookings recorded the highest number of cancellations, likely due to flexible modification or refund options provided by third-party travel websites.

### iii. FAMILY SIZE

- The average family size is 2 members, with most bookings made by couples or small families.
- Bookings with larger family sizes generally have lower cancellation rates, indicating stronger travel intent among group travelers.

### iv. CAR PARKING SPACE

- Most guests did not request any parking space, while a smaller portion booked one or more.
- Guests requesting parking facilities were more likely to confirm their stay, suggesting that those traveling by car are less likely to cancel.

### v. NUMBER OF WEEKEND NIGHTS

- The minimum number of weekend nights stayed is 0 and the maximum is 4, with an average of 1 night.
- This shows that most bookings were short weekend getaways, and cancellations

were slightly higher for shorter stays.

vi. NUMBER OF WEEK NIGHTS

- The minimum number of weekday nights is 0 and the maximum is 5, with an average of 2 nights.
- Longer weekday stays showed lower cancellation rates, indicating that extended visits are more stable and planned.

vii. TYPE OF MEAL

- Most guests selected Meal Plan 1 or Not Selected, with very few choosing Meal Plan 2.
- Since the variable showed very low variation and offered minimal predictive significance, it was not considered in the exploratory or modeling stages. The similarity in meal preferences across bookings suggests that meal plan choice has little to no influence on booking cancellations.

viii. ROOM TYPE

- Most bookings were for Room Type 1 and Room Type 4, indicating these are the most reserved categories.
- The variable largely reflects internal hotel room coding rather than customer decision-making behavior, and thus provided limited analytical value.

ix. BOOKING STATUS

- The dataset initially contained more “Not Canceled” bookings than “Canceled” ones.
- To address imbalance, the dataset was adjusted to ensure an approximately equal distribution, improving model accuracy. This step helped the decision tree classify both outcomes fairly without bias toward the majority group.

### 4.1.6.3 SUMMARY OF EDA INSIGHTS

The exploratory data analysis revealed clear patterns in customer booking behaviour. Online bookings and those with longer lead times were more likely to be cancelled, while shorter lead times and longer stays tended to have lower cancellation rates. Family size and car parking space were also found to influence cancellations — families and guests who requested parking showed stronger booking commitment compared to solo travellers.

Variables such as type of meal and room type were not included in the EDA since they displayed very low variance and offered minimal predictive value. Most guests selected the same meal plans or room categories, making these attributes statistically insignificant for the cancellation model.

Overall, the EDA helped identify the most relevant predictors, ensuring that only meaningful variables were retained for accurate and unbiased model development.

#### 4.1.6.4 CUSTOMER SEGMENTATION

Although customer segmentation was not a direct analytical objective of this study, the dataset inherently represents different types of guests based on their booking behaviour. Broadly, guests can be classified as:

- **Demographic Segment:** Based on family size, distinguishing solo travellers, couples, and families.
- **Behavioral Segment:** Based on lead time and market segment type, showing early online bookers versus last-minute direct customers.
- **Operational Segment:** Based on stay details like weekend/weekday nights and car parking space, reflecting travel intent and trip type.

This classification provides contextual understanding of guest profiles and complements the EDA findings, though no separate segmentation algorithm was applied.

#### 4.1.7 DATA ANALYSIS USING DECISION TREE

The Decision Tree algorithm was used to predict whether a hotel booking would be canceled or not canceled, based on various demographic, behavioral, and booking-related factors. The

model was developed and evaluated in R programming language, using a structured step-by-step analytical approach.

## STEP 1: SPLITTING THE DATASET INTO TRAINING AND TEST SETS

The first step in the process of analysis by using decision tree is splitting the data into training and test sets to ensure unbiased model evaluation.

- By using `set.seed(1234)`, you can ensure that the split is reproducible, resulting in the same training/test sets every time the code is run.
- The `createDataPartition()` function from the **caret** package was then used to randomly divide the data based on the target variable (booking status), while maintaining similar class proportions in both subsets.
- The training data is stored in 'train', while the test data is stored in 'test'.

The dataset is divided into training (80%) and testing (20%) sets to assess the model's performance. This split ensures that the model is trained on one portion of the data and evaluated on unseen data to test its predictive capability.

### R CODE

```
Split: create train and test sets (80% train, 20% test)
set.seed(123)
trainIndex <- createDataPartition(bookingdata1$`booking status`, p = 0.8, list = FALSE)
train <- bookingdata1[trainIndex,]
test <- bookingdata1[-trainIndex,]
```

**Figure 26**

*R code of splitting the dataset into training and test data*

### CODE EXPLANATION

`set.seed(1234)`

- sets the seed for R's random number generator to 1234.

```
trainIndex <- createDataPartition(bookingdata1$`booking status`, p = 0.8, list = FALSE)
```

- `createDataPartition()`: A function from the `caret` package that splits your data for model training and testing. Unlike random sampling, it keeps the same proportion of each class (Canceled / Not Canceled) in both sets. This is called stratified sampling.
- `p = 0.8`: Specifies that 80% of the data should go to the training set. The remaining 20% (automatically) goes to the test set.
- `list = FALSE`: Tells the function to return the row numbers (indices) directly, instead of putting them inside a list — which makes it easier to use for subsetting later.
- `trainIndex <-`: Stores those rows numbers in an object called `trainIndex`, which you'll use to extract the corresponding rows for your training data.

```
train <- bookingdata1[trainIndex,]
```

- creates the training dataset (`train`) by selecting the rows of `bookingdata1` identified in `trainIndex`, which will contain approx. 80% of the dataset 'bookingdata1'.

```
test <- bookingdata1[-trainIndex,]
```

- creates the test dataset (`test`) by selecting all remaining rows (those not in `trainIndex`), ensuring the model is evaluated on unseen data which will contain approx. 20% of the dataset 'bookingdata1'.

## RESULT

|         |                          |
|---------|--------------------------|
| ▶ test  | 2137 obs. of 9 variables |
| ▶ train | 8551 obs. of 9 variables |

**Figure 27**

*Screenshot of the result in R*

## STEP 2: VISUALISING DECISION TREE

This is the main step in analysis using decision tree in R programming language. It helps to create a new decision tree model using the balanced dataset. A Decision Tree model was developed using the `rpart()` function to predict booking cancellations based on multiple input variables. The tree structure helps visualize how different features, such as lead time, market segment type, and family size, contribute to the decision of whether a booking is canceled or not.

**Formula:**  $y \sim .$  indicates that  $y$  is the target variable (DV), and all the remaining columns are predictor variables (IVs).

**Data:** train

**Method:** "class" for classification tasks.

### R CODE

```
tree <- rpart(`booking status` ~ ., data = train,
 method = "class", control = rpart.control(maxdepth = 4))
rpart.plot(tree)
```

### Figure 28

*Rcode for building the decision tree*

### CODE EXPLANATION

```
tree <- rpart(`booking status` ~ ., data = train,
method = "class", control = rpart.control(maxdepth = 4))
```

- `rpart()` builds the decision tree model using all independent variables ( $\sim .$ ) to predict the target variable (booking status)
- `method = "class"` specifies that it is a classification task.
- `control = rpart.control(maxdepth = 4)` limits the tree's depth to prevent overfitting and improve interpretability.

```
rpart.plot(tree)
```

- visually displays the structure of the tree, showing key variables and decision splits that lead to each outcome.

### STEP 3: PREDICTION USING DECISION TREE MODEL

In this step, the fitted decision tree model (tree) is used to make predictions for the test dataset (test).

#### R CODE

```
pred_test <- predict(tree, newdata = test, type = "class")
```

#### Figure 29

*Rcode for predicting using the decision tree model*

#### CODE EXPLANATION

```
pred_test <- predict(tree, newdata = test, type = "class")
```

- The predict() function is utilized to generate forecasts using the model.
- The dataset for making predictions is indicated by newdata = test. This dataset includes the input features but not the target variable.
- “class” specifies that the class labels to be predicted – cancelled or not cancelled, rather than probabilities.

### STEP 4: EVALUATION OF MODEL USING CONFUSION MATRIX

This is the final step in the decision tree analysis process using R programming language. The assessment of decision tree model heavily relies on a confusion matrix. This matrix offers a concise overview of how the model's predictions align with the real outcomes. Essentially, it is a table that presents the accurate and inaccurate predictions made by the model.

## CONFUSION MATRIX COMPONENTS:

- TP (True Positives): The count of positive instances correctly predicted as positive.
- TN (True Negatives): The count of negative instances correctly predicted as negative.
- FP (False Positives): The count of negative instances incorrectly predicted as positive (also called Type I error).
- FN (False Negatives): The count of positive instances incorrectly predicted as negative (also called Type II error).

## R CODE

```
'conf_mat <- confusionMatrix(pred_test, test$`booking status`')
print(conf_mat)
```

### Figure 30

*R code of evaluation of model using confusion matrix*

## CODE EXPLANATION

```
confusionMatrix(data=bm1_predict, reference = bm1_testdt$y)
```

- confusionMatrix() (from the caret package) creates a summary table showing correct and incorrect predictions.
- pred\_test contains the predicted class labels, while test\$'booking status' holds the actual values.
- print(conf\_mat) displays key performance metrics such as Accuracy, Precision, Recall, and F1 Score, which indicate how well the model performed on unseen data.



## OUTPUT

```
Confusion Matrix and Statistics

Prediction Reference
 Canceled Not_Canceled
Canceled 960 378
Not_Canceled 118 681

 Accuracy : 0.7679
 95% CI : (0.7494, 0.7857)
No Information Rate : 0.5044
P-Value [Acc > NIR] : < 2.2e-16

 Kappa : 0.5348

McNemar's Test P-Value : < 2.2e-16

 Sensitivity : 0.8905
 Specificity : 0.6431
 Pos Pred Value : 0.7175
 Neg Pred Value : 0.8523
 Prevalence : 0.5044
 Detection Rate : 0.4492
 Detection Prevalence : 0.6261
 Balanced Accuracy : 0.7668

 'Positive' Class : Canceled
```

**Figure 31**

*Screenshot of confusion matrix in R*

## 2. DATA INTERPRETATION

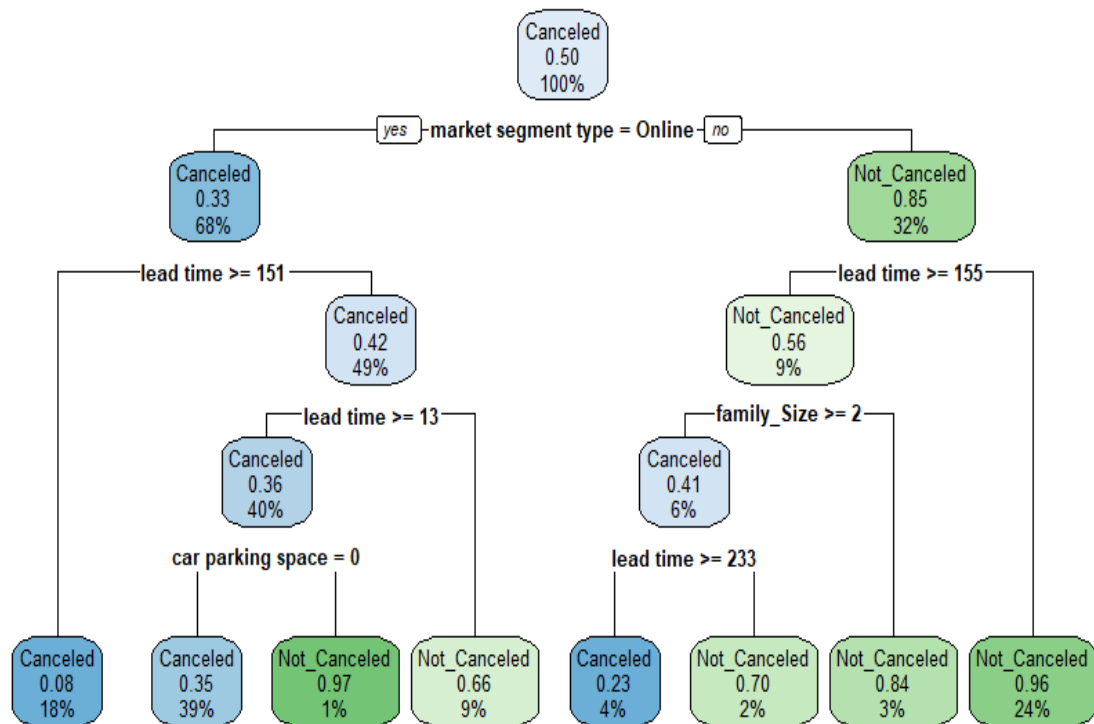
### DATA VISUALISATION: DECISION TREE USING TRAINING DATASET

The decision tree is visualized using the training dataset (train) in the STEP: VISUALISING DECISION TREE (see above). The visual representation of a decision tree includes nodes (in the form of boxes) and edges. A decision based on an input variable is depicted by each internal node, while the outcome of that decision is represented by each edge. The final predictions of the model are represented by the terminal nodes, also referred to as leaves.

The decision tree is visually represented in this plot to illustrate the decision-making process in the subscription of term deposit based on the marketing efforts.

## OUTPUT

The decision tree graph visualized is the final output (shown below).



**Figure 32**

*Decision tree using the training dataset*

### 4.2.1.1 INTERPRETATION

#### Overall Model Insights:

The decision tree visually represents how different features — particularly market segment type, lead time, family size, and car parking space — influence hotel booking cancellations. The root node shows that bookings are almost evenly split between Cancelled and Not Cancelled (50% each), suggesting that both outcomes are equally common in the dataset.

The tree begins by splitting on market segment type, indicating that this is the most significant variable influencing cancellation behaviour.

### **Key Interpretations:**

#### **1. Market Segment Type – The Primary Split**

- Guests who booked through online platforms have a much higher likelihood of cancellation, with 68% of bookings in this segment resulting in cancellations.
- In contrast, guests who booked offline or through direct/corporate channels have a strong tendency not to cancel, with 85% of those bookings being retained.
- This shows that online customers are more flexible and tend to cancel more often, while direct bookings are more stable and reliable.

#### **2. Lead Time – The Strongest Predictor After Segment Type**

- Among online bookings, lead time  $\geq 151$  days indicates a higher chance of cancellation (around 49% canceled).
- When lead time is short (less than 151 days), the chance of cancellation drops significantly, reflecting that last-minute bookings are more likely to be honored.
- Similarly, for offline bookings, lead time  $\geq 155$  days also leads to higher cancellations, showing that long gaps between booking and check-in increase the risk of cancellation.

#### **3. Car Parking Space and Family Size – Behavioral Indicators**

- For online customers with lead time  $\geq 13$  days, those who requested car parking space = 0 (no parking) showed a stronger tendency to cancel. This implies that guests not traveling by car may have less fixed travel plans or may be booking more speculatively.
- Conversely, bookings that included a car parking request were far more likely to be retained, indicating more concrete travel intent.
- In the offline segment, family size  $\geq 2$  acted as a stabilizing factor — larger families showed a higher likelihood of not canceling, suggesting group or family travel is usually well-planned

and less prone to changes.

#### **4. Combined Insights – Lead Time and Travel Intent**

- A combination of short lead time and family involvement yields the most stable bookings.
- On the other hand, online bookings with longer lead times and no parking request form the segment with the highest cancellation risk — reflecting spontaneous or flexible online travel planning behavior.

#### **Summary of Decision Tree Insights:**

- **High Cancellation Segments:** Online bookings with longer lead times and no parking request.
- **Low Cancellation Segments:** Offline or direct bookings, shorter lead times, and family or group stays.

### 4.4.1 CONFUSION MATRIX

In R, a confusion matrix is a tabular representation that compares predicted values against actual values. It consists of two dimensions, with one dimension showing the predicted values and the other dimension representing the actual values.

#### OUTPUT

The confusion matrix is depicted below in Figure 31.

```
Confusion Matrix and Statistics

 Prediction Reference
 Canceled Not_Canceled
Canceled 960 378
Not_Canceled 118 681

 Accuracy : 0.7679
 95% CI : (0.7494, 0.7857)
 No Information Rate : 0.5044
 P-Value [Acc > NIR] : < 2.2e-16

 Kappa : 0.5348

 Mcnemar's Test P-Value : < 2.2e-16

 Sensitivity : 0.8905
 Specificity : 0.6431
 Pos Pred Value : 0.7175
 Neg Pred Value : 0.8523
 Prevalence : 0.5044
 Detection Rate : 0.4492
 Detection Prevalence : 0.6261
 Balanced Accuracy : 0.7668

 'Positive' Class : Canceled
```

**Figure 33**  
*Screenshot of Confusion Matrix in R*

#### 4.4.1.1 INTERPRETATION

**Table 5**  
*Confusion Matrix*

|                             | <i>Predicted: Canceled</i> | <i>Predicted: Not Canceled</i> |
|-----------------------------|----------------------------|--------------------------------|
| <i>Actual: Canceled</i>     | 960 (TP) – True Positive   | 118 (FN) – False Negative      |
| <i>Actual: Not Canceled</i> | 378 (FP) – False Positive  | 681 (TN) – True Negative       |

| Metric                      | Interpretation                                                       |
|-----------------------------|----------------------------------------------------------------------|
| <b>True Positives (TP)</b>  | Correctly predicted canceled bookings (960 cases).                   |
| <b>False Positives (FP)</b> | Not-canceled bookings incorrectly predicted as canceled (378 cases). |
| <b>True Negatives (TN)</b>  | Correctly predicted not-canceled bookings (681 cases).               |
| <b>False Negatives (FN)</b> | Canceled bookings missed by the model (118 cases).                   |

Key metrics: Kappa, Accuracy, Precision, Recall and F1 score are metrics used to evaluate the performance of a classification mode

- Kappa:

Kappa values can range from -1 to 1, with 0 indicating random chance and 1 indicating perfect agreement between predictions and actual values. The Kappa statistic (0.5348) indicates a moderate level of agreement between the model's predictions and the actual outcomes. This suggests that the model performs substantially better than random guessing in predicting booking cancellations.

- Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$= 960+681/960+681+378+118$$

$$\text{Accuracy} = 0.7679$$

0.7679 means approximately 76.8 % of the model's predictions are correct. About three-quarters of all predictions correctly identify whether a booking was canceled or not.

- Precision

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{or} \quad \frac{\text{True Positive}}{\text{Predictive Results}}$$

$$= 960/960+378$$

$$\text{Precision} = 0.7175$$

Out of all the positive predictions we made, **71.75% is truly positive**. When the model predicts a booking will be canceled, it is correct roughly **72%** of the time. This shows good reliability in identifying true cancellations.

- Recall:

$$\text{Recall} = \frac{TP}{TP + FN} \text{ or } \frac{\text{True Positive}}{\text{Actual Results}}$$

$$= 960/960+118$$

$$\text{Recall} = 0.8905$$

Out of all the data points that should be predicted as true (actual results), **89% correctly predicted as true**. A recall of 89% shows that the model effectively captures most of the actual cancellations.

- Specificity:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$= 681/681+378$$

$$\text{Specificity} = 0.6431$$

The **specificity of 64.3%** implies that the model correctly identifies non-canceled bookings about two-thirds of the time. While lower than recall, it still provides reasonable discrimination.

- F1 Score

$$\text{F - score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}}$$

$$= 2*0.7175*0.8905/0.7175+0.8905$$

$$= 0.795$$

$$\text{F1 Score} = 0.795$$

F1 score is a single metric that is a harmonic mean of precision and recall. Here the F1 Score of **0.795**. It represents a balanced measure of the model's precision and recall, confirming consistent and dependable performance across both detection and accuracy dimensions.

**CHAPTER 5**  
**SUMMARY OF FINDINGS, SUGGESTIONS**  
**AND CONCLUSION**



## 5.1 SUMMARY OF FINDINGS

### 1. Influence of Lead Time:

The analysis revealed that lead time (the number of days between booking and arrival) plays a major role in predicting cancellations. Bookings with longer lead times were more likely to be canceled, as customers had more time to change their plans or find alternatives. Conversely, shorter lead times showed higher retention rates, suggesting that last-minute bookings are more reliable and involve confirmed travel intentions.

### 2. Impact of Market Segment:

The market segment type was found to be a key factor in determining cancellation behavior. Bookings made through online travel agencies (OTA) exhibited the highest cancellation rates, whereas direct or corporate bookings showed significantly lower rates. This indicates that customers booking online may be more price-sensitive or flexible, while direct customers are more committed to their plans.

### 5. Behavioral Trends and Customer Segments:

The decision tree highlighted that smaller families and solo travelers tend to cancel more frequently than group bookings. These insights can help hotels segment customers into risk categories and design appropriate engagement strategies, such as flexible offers or advance payment options for high-risk segments.

### 6. Model Performance and Predictive Strength:

The decision tree achieved an accuracy of 76.79%, indicating a strong overall performance. A recall of 89% showed that the model effectively identified most of the actual cancellations, while a precision of 72% reflected reliable predictive capability. The model's balanced accuracy of 76.68% and Kappa value of 0.53 suggest consistent and moderate agreement between actual and predicted outcomes.

## 5.2 SUGGESTIONS AND CONCLUSION

### 5.2.1 SUGGESTIONS

Based on the findings of this study, several actionable strategies can be proposed to reduce hotel booking cancellations and enhance operational efficiency:

1. Monitor and Manage Long Lead Time Bookings:

Since longer lead times increase the likelihood of cancellations, hotels should implement targeted reminders or partial prepayment options for such bookings. This can reduce the probability of guests canceling at the last minute.

2. Prioritize Direct and Corporate Channels:

Direct bookings tend to have lower cancellation rates. Therefore, hotels should strengthen loyalty programs, exclusive discounts, or value-added packages for customers booking directly through the hotel's website or sales team.

3. Design Family-Oriented Packages:

Families and guests requesting parking show higher commitment. Tailored family packages or special parking-inclusive deals can further encourage retention and reduce cancellations.

4. Targeted Retention Strategies for Online Bookings:

As online bookings show higher volatility, proactive communication (like confirmation calls, flexible rebooking options, or personalized offers) can be used to convert potential cancellations into retained bookings.

5. Utilize Predictive Analytics in Operations:

The model can be integrated into hotel management systems to flag "high-risk" bookings in real time, allowing staff to take preventive actions such as offering incentives or reconfirming stays before cancellation occurs.

## 5.2.2 PRACTICAL IMPLICATION

The findings of this study have strong practical relevance for the hospitality industry, particularly in hotel operations, marketing, and revenue management. The decision tree model provides actionable insights that can help hotels predict booking cancellations and take preventive measures to minimize financial and operational losses.

- ✓ **Enhanced Revenue Management:**

By identifying bookings with a high probability of cancellation, hotels can proactively adjust room availability and pricing strategies. This allows for dynamic overbooking management and helps maintain high occupancy rates without significant risk of revenue loss due to last-minute cancellations.

- ✓ **Operational Efficiency and Resource Planning:**

Predictive insights from the model enable hotels to allocate resources more effectively. Knowing which bookings are more likely to be canceled allows management to optimize staff scheduling, housekeeping, and inventory management, reducing unnecessary operational costs.

- ✓ **Targeted Customer Retention Strategies:**

The analysis highlights that online bookings and long lead times are more prone to cancellations. Hotels can use this information to implement targeted retention strategies, such as personalized confirmation emails, flexible cancellation policies, or small incentives for customers to retain their bookings.

- ✓ **Improved Marketing and Channel Optimization:**

Insights about market segment behavior can guide marketing teams to focus on channels with lower cancellation rates, such as direct and corporate bookings. Promotional efforts can be tailored to retain customers in segments that exhibit higher commitment, thereby improving conversion rates.

- ✓ **Customer Relationship Management (CRM):**

The decision tree findings can be integrated into CRM systems to flag customers at risk of canceling. This allows staff to engage with them in advance through follow-up calls, special

offers, or loyalty benefits, thereby improving guest satisfaction and long-term loyalty.

✓ **Data-Driven Decision Making:**

The project demonstrates how analytical models can assist hotel management in making strategic, evidence-based decisions. By continuously monitoring patterns in booking behavior, hotels can refine their policies and adapt to market dynamics more efficiently.

### **5.2.3 CONCLUSION**

This study on hotel booking cancellations provides valuable insights into the behavioral and operational factors influencing booking reliability. The findings revealed that market segment type and lead time are the most critical determinants of cancellation likelihood. Guests who booked online and far in advance were more prone to cancel, while direct, last-minute, and family bookings tended to remain confirmed.

The model's overall accuracy of 76.79% and high recall value underscore its effectiveness in predicting cancellations. This predictive capability enables hotels to take proactive measures, such as sending reminders, requesting partial payments, or offering flexible alternatives to at-risk customers.

Moreover, the decision tree's simplicity makes it an excellent decision-support tool for hotel managers, providing clear and interpretable rules for practical use. Implementing these insights can lead to reduced cancellations, improved room utilization, and enhanced customer satisfaction.

In conclusion, this project demonstrates how analytics and decision tree modelling can transform hotel operations by promoting smarter, data-backed decisions. By using predictive insights to anticipate customer behaviour, hotels can strengthen booking reliability, optimize revenue, and deliver more personalized guest experiences—ultimately contributing to a more efficient and resilient hospitality business.

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